

Q2)(a) Verification of $A=LU$

Observations: L is constructed from inverses of elementary matrices.

L_{ij} = factor means which we used $\in$ that subtracted factor $\times R_{-j}$ from R_{-i}

No. of elementary matrices used = 6

- L is built from inverses of elementary matrices (flip signs)

(b) Cholesky Decomposition

- Cholesky works for Symmetric positive Definite matrices
- It finds L such that $A = L \cdot L^T$
- It is more efficient w.r.t LU as we compute only lower triangle.
- $L_{ij} = (A_{ij} - \sum L_{ik} \cdot L_{jk}) / L_{ij}$
- SPD matrices ensure all diagonal elements remain positive.

(C) QR by Gram-Schmidt

- The QR decomposition successfully factorised 5×3 matrix into an orthogonal matrix Q and an upper triangular matrix R .
- The verification $Q^T Q = I$ confirms orthogonality of columns of Q .
- Non zero positive diagonal entries of R indicate linear independence of the columns of A .
- The reconstructing error is effectively 0 which confirms numerical correctness of the implementation.

(d) QR decomposition with Random 7×5 matrix
For given matrix A , Calculating diagonal elements of R
 $R = [13.1846, 9.8409, 11.6619, 6.3388, 8.0699]$

Observations:

- 1) All diagonal elements are positive.
- 2) This is the fundamental property of QR decomposition when columns are linearly independent.
- 3) We were able to verify $A = QR$, Max diff = 0.0
- 4) diagonal elements R_{jj} reveal column relationships.
- 5) Positive diagonal means columns are linearly independent.
- 6) each $R_{jj} = \|\text{residual}\|$ after removing projection onto previous columns.